MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Anubhav Mishra
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Education

MPSTME, NMIMS University **2023 – Present**
B.Tech in Computer Science
CGPA: 3.76 / 4 (Current Sem)

Shri Ramkrishna Harikrishna Academy, Surat **2023**
Senior Secondary (CBSE): 88.6 %

Shri Ramkrishna Harikrishna Academy, Surat **2021**
Secondary (CBSE) — 91%

Projects

Collections AI Agent – Simulated Loan Collections System (2026)

- Built an LLM collections agent using Gemini function calling, FastAPI, and MongoDB on fictional data, enforcing verification, lender plan rules, and human handoff in server code rather than prompts.
- Developed a 38-scenario evaluation suite covering prompt injection, wrong-borrower access, and invented policy, raising passes from 37/38 to 38/38 with 0 unauthorized actions in both runs.
- Implemented RAG for policy answers using local ONNX embeddings and NumPy vector search, with lender filtering, a score threshold, and source citations.
- Benchmarked regex and LLM promise extraction on 24 held-out fictional transcripts, achieving date F1 scores of 96.8% and 100%, and 97% F1 for both on promise and amount detection.
- Automated a nightly bash and PowerShell pipeline that extracts promises, loads SQLite, and runs SQL reports using CTEs and window functions.
- **Tech Stack:** Python, FastAPI, Gemini, MongoDB, SQLite, fastembed, NumPy, React, TypeScript, pytest

LogiSketch – AI-powered Diagram Generator (2025)

- Built an AI-powered diagram generation backend supporting 15 engines, using Express.js, Gemini, and Kroki to generate ER, flowchart, sequence, UML, and architecture diagrams.
- Implemented RAG over engine-specific syntax notes using local retrieval, raising Kroki render success from 88.9% to 97.8% (40/45 to 44/45) on a fixed test set.
- Built an evaluation pipeline for 45 prompts across 15 engines, comparing RAG on and off using code-scored render success, required-element coverage, latency, and token usage.
- Secured APIs using JWT authentication, request validation, Arcjet rate limiting, and centralized error handling.
- **Tech Stack:** Node.js, Express.js, MongoDB, Mongoose, Gemini, Kroki API

NodeNotion – Full Stack Blogging Platform (2025)

- Developed a full-stack blogging platform with markdown publishing, an admin dashboard, and content moderation.
- Secured role-based access using JWT authentication, bcryptjs hashing, and Firebase OAuth.
- **Tech Stack:** MongoDB, Express.js, React.js, Node.js, Redux Toolkit, Firebase, Supabase

Publications

- Published research paper: “A Multi-Factor Analysis of Gaming Habits and Their Impact on Mental Health”, IEEE Xplore; presented at IEEE DELCON 2025

Core Skills

- Languages: C++, Python, SQL, JavaScript
- AI/LLM: LLM Integration, Function Calling, RAG, Embeddings, Prompt Engineering, LLM Evaluation, Entity Extraction
- Frontend: HTML5, CSS3, React.js, Redux Toolkit, Tailwind CSS
- Backend: FastAPI, Node.js, Express.js
- Database: MongoDB, MySQL
- Tools: Git, GitHub, pytest, Postman, VS Code, Firebase, Supabase, Mongoose

Achievements

- Secured Team Rank 123 among 19,000+ participants in Round 1 of Odoo Hackathon 2025.
- Ranked among the Top 50 of 150+ teams in Yugantar Hackathon with "DeepSeak", an AI-based eDNA analysis pipeline.